FINITE-HEIGHT ATOMIC AND EXPONENTIAL-POLYNOMIAL CRITERIA FOR THE RIEMANN HYPOTHESIS

ABSTRACT. For a fixed height H with $N(H) > 0$, two normalized finite sums are associated with the nontrivial zeros $\rho = \beta + i\gamma$ of the Riemann zeta function:

$$L_H(x) = \frac{1}{N(H)} \sum_{0<\gamma\le H} x^{\rho-1/2}, \qquad M_H(x) = \frac{1}{N(H)} \sum_{0<\gamma\le H} x^{i\gamma}. \tag{0.1}$$

The first retains the horizontal coordinates of the zeros, whereas the second retains only their ordinates. We prove an exact decomposition of $L_H - M_H$ into positive defects localized at the distinct ordinates. After the logarithmic change of variable $x = e^y$, the difference is shown to be a finite exponential polynomial with constant complex exponents and constant coefficients. This yields three finite-height equivalences: equality on a set with an accumulation point, equality of the associated atomic measures at one fixed $x \neq 1$, and vanishing of a nonnegative second derivative at $y = 0$ are each equivalent to every zero up to height H lying on the critical line. Consequently, each criterion along any unbounded sequence of heights is equivalent to the Riemann Hypothesis. The finite-height arguments use only the functional-equation symmetry of the zeros, finiteness below a fixed height, positivity, and uniqueness of finite exponential polynomials; no formula of Landau type is used. In a logically separate triadic arithmetic section we prove an iterable sesquilinear contraction on every fixed active shell by using a fresh pair of primes at each level. The exact channel identities yield finite-depth quantitative estimates but do not supply an identity connecting the arithmetic channels to the horizontal zero coordinates.

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1. Introduction

Let $\zeta(s)$ denote the Riemann zeta function. Its nontrivial zeros lie in the critical strip $0 < \Re s < 1$, occur with finite multiplicity, and satisfy the symmetries

$$\rho \longmapsto \overline{\rho}, \qquad \rho \longmapsto 1-\rho, \qquad \rho \longmapsto 1-\overline{\rho}, \tag{1.1}$$

which follow from the functional equation and conjugation; see, for example, [4, 2]. In particular, if

$$\rho = \frac{1}{2} + \delta + i\tau \tag{1.2}$$

with $\tau > 0$, then

$$1-\overline{\rho} = \frac{1}{2} - \delta + i\tau \tag{1.3}$$

occurs with the same multiplicity.

There are no nontrivial real zeros: in particular,

$$\zeta(\sigma) \neq 0 \qquad (0 < \sigma < 1). \tag{1.4}$$

Thus every nontrivial zero has nonzero ordinate. This standard fact will be used explicitly when the positive-ordinate criteria are promoted to global formulations; see [4, 2].

Fix a height $H > 0$ with $N(H) > 0$. We compare two finite objects. The complete-zero sum weights a zero by its full displacement from the critical line, whereas the ordinate-only sum discards that horizontal displacement. The central question is whether equality of these two finite functions forces the zeros under consideration onto the critical line.

The answer is affirmative. The key point is not merely that the difference is analytic. After the substitution $x = e^y$, it becomes a finite sum of exponentials $e^{\lambda y}$ with constant complex exponents λ and constant coefficients. The apparent y-dependence of the ordinate-local coefficients disappears after expanding the hyperbolic cosine. Uniqueness of exponential polynomials then rules out every off-line exponent.

A second argument is even more local. The functional-equation symmetry makes the value and first derivative of the difference vanish automatically at $y = 0$. Its second derivative is

$$\frac{2}{N(H)} \sum_{\substack{\rho=\beta+i\gamma \\ 0<\gamma\le H \\ \beta>1/2}} \left(\beta - \frac{1}{2}\right)^2, \tag{1.5}$$

a nonnegative finite sum. Thus a single second-order equality at $x = 1$ already characterizes the finite-height Riemann Hypothesis.

The finite-height part is self-contained apart from standard facts about the zero symmetries and the identity theorem for holomorphic functions. It uses no explicit formula, prime sum, zero-density estimate, pair-correlation result, or Landau–Gonek formula. The later arithmetic part is separate and invokes Tao's proved logarithmically averaged two-point Elliott theorem only to certify orthogonality between distinct fixed arithmetic branches.

2. Finite zero sets and normalized transforms

Definition 2.1 (Finite zero multiset [Standard])**.** For $H > 0$, let

$$\mathcal{Z}_H = \{\rho = \beta + i\gamma : \zeta(\rho) = 0,\ 0 < \gamma \le H\} \tag{2.1}$$

be the multiset of nontrivial zeros with positive ordinate at most H, each repeated according to its multiplicity. Define

$$N(H) = |\mathcal{Z}_H|. \tag{2.2}$$

Remark 2.2. The multiset $\mathcal{Z}_H$ is finite for every fixed H because the zeros of a nonzero meromorphic function are isolated and a compact rectangle contains only finitely many of them. The classical Riemann–von Mangoldt formula provides a much sharper count, but no asymptotic counting formula is needed here [4].

Definition 2.3 (Complete and ordinate-only transforms [New formulation, unconditional])**.** Assume $N(H) > 0$. For $x > 0$, define

$$L_H(x) = \frac{1}{N(H)} \sum_{\rho=\beta+i\gamma\in\mathcal{Z}_H} x^{\rho-1/2}, \tag{2.3}$$

$$M_H(x) = \frac{1}{N(H)} \sum_{\rho=\beta+i\gamma\in\mathcal{Z}_H} x^{i\gamma}. \tag{2.4}$$

The powers are defined using the real logarithm,

$$x^z = e^{z\log x} \qquad (x > 0,\ z \in \mathbb{C}). \tag{2.5}$$

Remark 2.4. Assume $N(H) > 0$. The sum defining M_H repeats $x^{i\gamma}$ once for every zero at ordinate γ. Equivalently, if $m(\tau)$ denotes the total multiplicity at a distinct ordinate τ, then

$$M_H(x) = \frac{1}{N(H)} \sum_{0<\tau\le H} m(\tau) x^{i\tau}. \tag{2.6}$$

Definition 2.5 (Finite-height Riemann property [Definition])**.** We write $\mathrm{RH}(H)$ for the assertion

$$\Re\rho = \frac{1}{2} \qquad \text{for every } \rho \in \mathcal{Z}_H. \tag{2.7}$$

3. The exact ordinate-local defect

Definition 3.1 (Horizontal defect at one ordinate [New formulation, unconditional])**.** For a distinct ordinate $0 < \tau \le H$ and $x > 0$, define

$$D_x(\tau) = \sum_{\substack{\rho=\beta+i\tau\in\mathcal{Z}_H \\ \beta>1/2}} \left[\cosh\left(\left(\beta - \frac{1}{2}\right)\log x\right) - 1\right], \tag{3.1}$$

where multiplicities are retained.

Proposition 3.2 (Positivity and local vanishing [Unconditional])**.** *For every $x > 0$ and $0 < \tau \le H$,*

$$D_x(\tau) \ge 0. \tag{3.2}$$

If $x \ne 1$, then

$$D_x(\tau) = 0 \tag{3.3}$$

if and only if every zero at ordinate τ lies on the critical line.

Proof. Every summand in (3.1) is nonnegative because

$$\cosh u - 1 \ge 0 \qquad (u \in \mathbb{R}), \tag{3.4}$$

with equality if and only if $u = 0$. If $x \ne 1$, then $\log x \ne 0$. Hence a summand vanishes exactly when $\beta - 1/2 = 0$. The functional-equation symmetry pairs every zero with $\beta < 1/2$ to one with $\beta > 1/2$ at the same ordinate and with the same multiplicity. Therefore the absence of terms with $\beta > 1/2$ is equivalent to all zeros at that ordinate satisfying $\beta = 1/2$. □

Theorem 3.3 (Exact finite-height defect identity [New theorem, unconditional])**.** *Assume $N(H) > 0$. For every $x > 0$,*

$$\boxed{L_H(x) - M_H(x) = \frac{2}{N(H)} \sum_{0<\tau\le H} D_x(\tau) e^{i\tau \log x}.} \tag{3.5}$$

The outer sum is over distinct positive ordinates.

Proof. Starting from Definitions 2.3, group the terms by their distinct ordinates:

$$L_H(x) - M_H(x) = \frac{1}{N(H)} \sum_{0<\tau\le H} e^{i\tau \log x} \sum_{\rho=\beta+i\tau} \left(x^{\beta-1/2} - 1\right). \tag{3.6}$$

Fix τ. Zeros on the critical line contribute zero. For an off-line pair

$$\frac{1}{2} + \delta + i\tau, \qquad \frac{1}{2} - \delta + i\tau, \qquad \delta > 0, \tag{3.7}$$

the contribution to the inner sum in (3.6) is

$$x^{\delta} - 1 + x^{-\delta} - 1 = 2\cosh(\delta \log x) - 2 \tag{3.8}$$

$$= 2\left[\cosh(\delta \log x) - 1\right]. \tag{3.9}$$

Summing over all off-line pairs with multiplicity gives

$$\sum_{\rho=\beta+i\tau} \left(x^{\beta-1/2} - 1\right) = 2D_x(\tau). \tag{3.10}$$

Substitution of (3.10) into (3.6) proves (3.5). □

Remark 3.4. The representation (3.5) must not be used directly as a constant-coefficient exponential polynomial in the variable $y = \log x$, because $D_{e^y}(\tau)$ depends on y. The constant-coefficient form is obtained only after expanding every hyperbolic cosine, as carried out in Section 5.

4. Atomic measure formulation

Definition 4.1 (Complete and ordinate-only atomic measures [New formulation, unconditional])**.** For fixed H with $N(H) > 0$ and $x > 0$, define

$$\nu_{H,x} = \frac{1}{N(H)} \sum_{0<\tau\le H} \left(\sum_{\rho=\beta+i\tau} x^{\beta-1/2} \right) e^{i\tau \log x} \delta_\tau, \tag{4.1}$$

$$\mu_{H,x} = \frac{1}{N(H)} \sum_{0<\tau\le H} m(\tau) e^{i\tau \log x} \delta_\tau. \tag{4.2}$$

Theorem 4.2 (Positive atomic difference [New theorem, unconditional])**.** *Assume $N(H) > 0$. For every $x > 0$,*

$$\boxed{\frac{1}{2} e^{-i(\log x)t} \left(\nu_{H,x} - \mu_{H,x}\right) = \frac{1}{N(H)} \sum_{0<\tau\le H} D_x(\tau) \delta_\tau,} \tag{4.3}$$

where multiplication by $e^{-i(\log x)t}$ is understood atomwise. The measure on the right is nonnegative.

Proof. At an ordinate τ, the coefficient of $\nu_{H,x} - \mu_{H,x}$ is

$$\frac{e^{i\tau \log x}}{N(H)} \sum_{\rho=\beta+i\tau} \left(x^{\beta-1/2} - 1\right). \tag{4.4}$$

By (3.10), this equals

$$\frac{2e^{i\tau \log x}}{N(H)} D_x(\tau). \tag{4.5}$$

Demodulation and division by two give (4.3). Positivity follows from Proposition 3.2. □

Corollary 4.3 (One-parameter atomic criterion [New theorem, unconditional])**.** *Let $H > 0$ with $N(H) > 0$. For any fixed $x > 0$ with $x \neq 1$,*

$$\boxed{\nu_{H,x} = \mu_{H,x} \quad \Longleftrightarrow \quad \mathrm{RH}(H).} \tag{4.6}$$

Proof. If $\mathrm{RH}(H)$ holds, then every weight $x^{\beta-1/2}$ equals one, so the measures coincide. Conversely, equality of the measures makes the positive measure on the right of (4.3) zero. Every mass $D_x(\tau)$ therefore vanishes. Proposition 3.2 gives $\mathrm{RH}(H)$. □

Remark 4.4 (No atom-separating windows are required). The measures in Definition 4.1 have finite support on distinct ordinates. Equality of two such atomic measures is equality of the coefficient at each atom. Equivalently, the nonnegative measure on the right of (4.3) is zero if and only if every one of its atomic masses is zero. No auxiliary family of disjoint windows separating the ordinates is required.

5. Expansion as a finite exponential polynomial

Definition 5.1. Let $H > 0$ with $N(H) > 0$. For $y \in \mathbb{R}$, define

$$F_H(y) = L_H(e^y) - M_H(e^y). \tag{5.1}$$

Proposition 5.2 (Constant-coefficient expansion [New theorem, unconditional])**.** *Assume $N(H) > 0$. The function F_H extends to an entire function of $y \in \mathbb{C}$ and admits the finite expansion*

$$F_H(y) = \frac{1}{N(H)} \sum_{\rho=\beta+i\gamma\in\mathcal{Z}_H} \left[e^{(\beta-1/2+i\gamma)y} - e^{i\gamma y}\right]. \tag{5.2}$$

Equivalently, after grouping symmetric pairs,

$$F_H(y) = \frac{1}{N(H)} \sum_{0<\tau\le H} \sum_{\delta>0} m_H(\delta,\tau) \left[e^{(\delta+i\tau)y} + e^{(-\delta+i\tau)y} - 2e^{i\tau y}\right], \tag{5.3}$$

where $m_H(\delta,\tau)$ is the multiplicity of the zero $1/2+\delta+i\tau$ in $\mathcal{Z}_H$. Both exponents and coefficients in (5.3) *are constant.*

Proof. Substituting $x = e^y$ into (2.3) and (2.4) gives

$$L_H(e^y) = \frac{1}{N(H)} \sum_{\rho=\beta+i\gamma\in\mathcal{Z}_H} e^{(\beta-1/2+i\gamma)y}, \tag{5.4}$$

$$M_H(e^y) = \frac{1}{N(H)} \sum_{\rho=\beta+i\gamma\in\mathcal{Z}_H} e^{i\gamma y}. \tag{5.5}$$

Their difference is (5.2). It is entire because it is a finite sum of entire exponentials.

For a pair (3.7), the corresponding part of (5.2) is

$$e^{(\delta+i\tau)y} + e^{(-\delta+i\tau)y} - 2e^{i\tau y}. \tag{5.6}$$

Summing these terms with multiplicity gives (5.3). No y-dependent coefficient remains. □

Remark 5.3. The factors $e^{(\beta-1/2)y} - 1$ and $D_{e^y}(\tau)$ are indeed functions of y. This does not obstruct the uniqueness argument because they are not retained as coefficients. Expanding them produces the constant-coefficient expression (5.3).

6. Uniqueness of finite exponential polynomials

The following standard lemma is included with proof. It is a finite-dimensional form of uniqueness for exponential polynomials; related general theory may be found in [1].

Lemma 6.1 (Uniqueness lemma [Classical, unconditional]). *Let $\lambda_1, \dots, \lambda_r \in \mathbb{C}$ be distinct and let $a_1, \dots, a_r \in \mathbb{C}$. If*

$$G(y) = \sum_{j=1}^{r} a_j e^{\lambda_j y} \tag{6.1}$$

vanishes on a nonempty open interval of the real axis, then

$$a_1 = \cdots = a_r = 0. \tag{6.2}$$

Proof. The function G is entire. Since its zero set contains a real interval and therefore has an accumulation point in $\mathbb{C}$, the identity theorem gives

$$G(y) = 0 \qquad (y \in \mathbb{C}). \tag{6.3}$$

Fix $y_0 \in \mathbb{C}$. Differentiating (6.3) for $k = 0, \dots, r-1$ yields

$$\sum_{j=1}^{r} a_j e^{\lambda_j y_0} \lambda_j^k = 0. \tag{6.4}$$

This is a linear system for the quantities $a_j e^{\lambda_j y_0}$ with Vandermonde matrix

$$V = (\lambda_j^k)_{0 \le k \le r-1,\ 1 \le j \le r}. \tag{6.5}$$

Its determinant is

$$\det V = \prod_{1 \le j < k \le r} (\lambda_k - \lambda_j), \tag{6.6}$$

which is nonzero because the exponents are distinct. Hence

$$a_j e^{\lambda_j y_0} = 0 \qquad (1 \le j \le r). \tag{6.7}$$

Since exponentials never vanish, (6.2) follows. □

7. The finite-height interval criterion

Theorem 7.1 (Finite-height interval criterion [New theorem, unconditional]). *Let $H > 0$ with $N(H) > 0$. The following assertions are equivalent:*

(i) RH(H) *holds.*
(ii) $L_H(x) = M_H(x)$ *for every* $x > 0$.
(iii) *There exists a nonempty open interval* $J \subset (0, \infty)$ *such that*

$$L_H(x) = M_H(x) \qquad (x \in J). \tag{7.1}$$

(iv) *There exists a subset* $E \subset (0, \infty)$ *having an accumulation point in* $(0, \infty)$ *such that*

$$L_H(x) = M_H(x) \qquad (x \in E). \tag{7.2}$$

Proof. Assume (i). Then every zero in $\mathcal{Z}_H$ satisfies $\beta = 1/2$. Consequently,

$$x^{\rho - 1/2} = x^{i\gamma} \tag{7.3}$$

for every $x > 0$, and summation gives (ii). The implications (ii)⇒(iii)⇒(iv) are immediate.

Assume (iv). Under the homeomorphism $y = \log x$, the set

$$\log E = \{\log x : x \in E\} \tag{7.4}$$

has an accumulation point on the real axis. By (5.1),

$$F_H(y) = 0 \qquad (y \in \log E). \tag{7.5}$$

Proposition 5.2 shows that F_H is entire, so the identity theorem gives

$$F_H(y) \equiv 0. \tag{7.6}$$

Suppose, for contradiction, that RH(H) fails. Then there is a zero

$$\rho_0 = \frac{1}{2} + \delta_0 + i\tau_0 \tag{7.7}$$

with $\delta_0 > 0$. In the grouped expansion (5.3), the exponential

$$e^{(\delta_0 + i\tau_0)y} \tag{7.8}$$

has coefficient

$$\frac{m_H(\delta_0, \tau_0)}{N(H)} > 0. \tag{7.9}$$

No negative term in (5.3) has the same exponent. Indeed, an exponent $-\delta + i\tau$ has negative real part, an exponent $i\tau$ has real part zero, and another exponent $\delta + i\tau$ agrees with $\delta_0 + i\tau_0$ only when both parameters agree, in which case the multiplicities add positively. After grouping equal exponents, the coefficient in (7.9) therefore remains nonzero.

This contradicts Lemma 6.1 applied to (7.6). Hence RH(H) holds, proving (iv)⇒(i). □

Remark 7.2. Assume $N(H) > 0$. Equality at the single point $x = 1$ is automatic:

$$L_H(1) = M_H(1) = 1. \tag{7.10}$$

It contains no information about the horizontal coordinates. The interval or accumulation-point hypothesis in Theorem 7.1 is what activates analytic uniqueness.

8. A sharper second-derivative criterion

The interval criterion can be strengthened substantially. The second derivative at the single logarithmic point $y = 0$ already detects every off-line pair through a positive sum.

Proposition 8.1 (Automatic zeroth and first order agreement [Unconditional])**.** *Assume $N(H) > 0$. The function F_H satisfies*

$$F_H(0) = 0, \tag{8.1}$$

$$F_H'(0) = 0. \tag{8.2}$$

Proof. Equation (8.1) follows from (7.10). Differentiating (5.2) gives

$$F_H'(0) = \frac{1}{N(H)} \sum_{\rho = \beta + i\gamma \in \mathcal{Z}_H} \left(\beta - \frac{1}{2}\right). \tag{8.3}$$

At each fixed ordinate, the functional-equation symmetry pairs $\beta - 1/2$ with its negative and preserves multiplicity. The sum is therefore zero. □

Theorem 8.2 (Positive curvature identity [New theorem, unconditional])**.** *Assume $N(H) > 0$. The logarithmic second derivative of the defect is*

$$\boxed{F_H''(0) = \frac{2}{N(H)} \sum_{\substack{0 < \tau \le H \\ \delta > 0}} m_H(\delta, \tau)\delta^2 = \frac{2}{N(H)} \sum_{\substack{\rho = \beta + i\gamma \in \mathcal{Z}_H \\ \beta > 1/2}} \left(\beta - \frac{1}{2}\right)^2.} \tag{8.4}$$

In particular, $F_H''(0)$ is real and nonnegative.

Proof. Differentiate the contribution of one symmetric pair in (5.3) twice. At $y = 0$ it gives

$$(\delta + i\tau)^2 + (-\delta + i\tau)^2 - 2(i\tau)^2 \tag{8.5}$$

$$= \left(\delta^2 + 2i\delta\tau - \tau^2\right) + \left(\delta^2 - 2i\delta\tau - \tau^2\right) + 2\tau^2 \tag{8.6}$$

$$= 2\delta^2. \tag{8.7}$$

Summation over all pairs with multiplicity and division by $N(H)$ yields (8.4). □

Corollary 8.3 (Finite-height curvature criterion [New theorem, unconditional])**.** *Let $H > 0$ with $N(H) > 0$. The following assertions are equivalent:*

$$\mathrm{RH}(H), \tag{8.8}$$

$$F_H''(0) = 0. \tag{8.9}$$

Equivalently,

$$\boxed{\left.\frac{\mathrm{d}^2}{\mathrm{d}y^2} L_H(e^y)\right|_{y=0} = \left.\frac{\mathrm{d}^2}{\mathrm{d}y^2} M_H(e^y)\right|_{y=0} \iff \mathrm{RH}(H).} \tag{8.10}$$

Proof. If $\mathrm{RH}(H)$ holds, the sum in (8.4) is empty. Conversely, every term on the right of (8.4) is nonnegative. If the sum vanishes, then no $\delta > 0$ occurs. Functional-equation symmetry then places every zero in $\mathcal{Z}_H$ on the critical line. □

Corollary 8.4 (Euler-operator form [Unconditional])**.** *Let $H > 0$ with $N(H) > 0$. Let*

$$\mathcal{E} = x\frac{\mathrm{d}}{\mathrm{d}x}. \tag{8.11}$$

Then

$$\boxed{\mathcal{E}^2\big(L_H(x) - M_H(x)\big)\Big|_{x=1} = \frac{2}{N(H)} \sum_{\substack{\rho \in \mathcal{Z}_H \\ \Re\rho > 1/2}} \left(\Re\rho - \frac{1}{2}\right)^2.} \tag{8.12}$$

Consequently, the left side vanishes if and only if $\mathrm{RH}(H)$ *holds.*

Proof. Under $x = e^y$, the operator $\mathrm{d}/\mathrm{d}y$ equals $x\,\mathrm{d}/\mathrm{d}x = \mathcal{E}$. Apply Theorem 8.2. □

9. Global formulations equivalent to the Riemann Hypothesis

Theorem 9.1 (Unbounded-height interval criterion [New theorem, unconditional])**.** *The Riemann Hypothesis is equivalent to the existence of an unbounded sequence $H_j \to \infty$, with $N(H_j) > 0$, such that for each j there exists a subset $E_j \subset (0, \infty)$ having an accumulation point in $(0, \infty)$ and*

$$L_{H_j}(x) = M_{H_j}(x) \qquad (x \in E_j). \tag{9.1}$$

Proof. If RH holds, then $\mathrm{RH}(H_j)$ holds for every H_j, and Theorem 7.1 gives equality for every $x > 0$.

Conversely, suppose the stated condition holds. Let $\rho = \beta + i\gamma$ be any nontrivial zero with $\gamma > 0$. Choose j such that $H_j \geq \gamma$. Theorem 7.1 applied at height H_j gives $\mathrm{RH}(H_j)$, so $\beta = 1/2$. Conjugation then treats zeros with negative ordinate as well. Equation (1.4) excludes a remaining zero with ordinate zero. Thus every nontrivial zero lies on the critical line. □

Theorem 9.2 (Unbounded-height curvature criterion [New theorem, unconditional])**.** *The Riemann Hypothesis is equivalent to the existence of an unbounded sequence $H_j \to \infty$, with $N(H_j) > 0$, such that*

$$F_{H_j}''(0) = 0 \qquad (j \geq 1). \tag{9.2}$$

Proof. Under RH, Corollary 8.3 gives (9.2) for every height with $N(H) > 0$. Conversely, each equality in (9.2) implies $\mathrm{RH}(H_j)$. Since the sequence is unbounded, every zero with positive ordinate is eventually included. Conjugation treats zeros with negative ordinate, and (1.4) excludes nontrivial zeros with ordinate zero. □

Theorem 9.3 (Unbounded-height atomic criterion [New theorem, unconditional])**.** *Fix any $x_0 > 0$ with $x_0 \neq 1$. The Riemann Hypothesis is equivalent to the existence of an unbounded sequence $H_j \to \infty$, with $N(H_j) > 0$, such that*

$$\nu_{H_j,x_0} = \mu_{H_j,x_0} \qquad (j \geq 1). \tag{9.3}$$

Proof. Apply Corollary 4.3 at every H_j. The resulting finite-height assertions exhaust all zeros with positive ordinate. Conjugation treats zeros with negative ordinate, and (1.4) excludes nontrivial zeros with ordinate zero. □

10. Logical scope and remaining problem

Proposition 10.1 (What has been proved [Unconditional])**.** *For every finite height H with $N(H) > 0$, each of the following is an exact criterion for* $\mathrm{RH}(H)$*:*

(a) *equality of L_H and M_H on any set with an interior accumulation point;*
(b) *equality of the complete and ordinate-only atomic measures at any fixed $x \neq 1$;*
(c) *vanishing of the positive curvature $F''_H(0)$.*

Along any unbounded sequence of heights, each criterion is equivalent to the full Riemann Hypothesis.

Proof. The three finite-height assertions are Theorem 7.1, Corollary 4.3, and Corollary 8.3. Their global versions are Theorems 9.1, 9.3, and 9.2. □

10.1. **Normalized convergence does not activate the criteria.**

Proposition 10.2 (Model obstruction to normalized convergence [Unconditional])**.** *The identity* (8.4) *has the following precise limitation: the convergence*

$$F''_{H_j}(0) \longrightarrow 0 \qquad (j \longrightarrow \infty) \tag{10.1}$$

can occur for sequences of symmetric finite multisets for which a fixed off-line pair persists. Thus the identity controls an average, whereas the finite-height criteria require exact vanishing at each relevant height.

Proof. Identity (8.4) is the normalized nonnegative moment

$$F''_{H_j}(0) = \frac{2}{N(H_j)} \sum_{\substack{\rho \in \mathcal{Z}_{H_j} \\ \Re\rho > 1/2}} \left(\Re\rho - \frac{1}{2}\right)^2. \tag{10.2}$$

From this identity alone, a fixed off-line pair with horizontal displacement $\delta_0 > 0$ contributes only

$$\frac{2\delta_0^2}{N(H_j)}, \tag{10.3}$$

which tends to zero as $N(H_j) \to \infty$. More formally, symmetric finite multisets consisting of one fixed off-line pair and an increasing number of critical-line zeros have normalized curvature tending to zero while failing the corresponding finite-height critical-line property at every stage. This model proves the stated limitation of the normalized identity; it does not assert the existence of such a zero configuration for the Riemann zeta function. Exact vanishing, or an additional estimate preventing such dilution, is indispensable. □

10.2. **A finite triadic contraction and its precise obstruction.** The following calculation records the one-step contraction of a linear prime average and its invariant component.

Proposition 10.3 (Finite triadic averaging [Unconditional])**.** *Let $j \geq 2$, $q = 3^j$, $U_j = (\mathbb{Z}/q\mathbb{Z})^\times$, and $\zeta_q = e^{2\pi i/q}$. Define*

$$c_j(u) = -\frac{3}{2}(-1)^{j-1}3^{-j}(1+\zeta_q^u) \qquad (u \in U_j), \tag{10.4}$$

and let

$$(\Pi_{a,j}T)(u) = T(au) \qquad (a \in U_j). \tag{10.5}$$

All inner products and norms in this subsection are taken in $\ell^2(U_j)$ with counting measure and with the convention

$$\langle f, g\rangle_{\ell^2(U_j)} = \sum_{u\in U_j} f(u)\overline{g(u)}. \tag{10.6}$$

With this convention,

$$\Pi_{a,j}^* = \Pi_{a^{-1},j}. \tag{10.7}$$

For the two-prime average

$$\mathcal{A}_j = \frac{1}{2}(\Pi_{5,j} + \Pi_{7,j}), \tag{10.8}$$

one has

$$\|\mathcal{A}_j^* c_j\|_2^2 = \frac{3}{4}\|c_j\|_2^2. \tag{10.9}$$

Nevertheless,

$$\|(\mathcal{A}_j^*)^k c_j\|_2 \geq \frac{1}{\sqrt{2}}\|c_j\|_2 \qquad (k \geq 1), \tag{10.10}$$

and the pointwise inequality

$$|\langle \mathcal{A}_j T, c_j\rangle| < |\langle T, c_j\rangle| \tag{10.11}$$

does not hold for arbitrary channel vectors T.

Proof. Write

$$\alpha_j = -\frac{3}{2}(-1)^{j-1}3^{-j}. \tag{10.12}$$

For units a, b modulo q, expansion of (10.4) gives

$$\langle \Pi_{a,j}^* c_j, \Pi_{b,j}^* c_j\rangle = |\alpha_j|^2(\varphi(q) + r_q(a^{-1} - b^{-1})), \tag{10.13}$$

where

$$r_q(d) = \sum_{u\in U_j} \zeta_q^{du} \tag{10.14}$$

is the Ramanujan sum. Since

$$v_3(5^{-1} - 7^{-1}) = v_3(7-5) = 0, \tag{10.15}$$

one has $r_{3^j}(5^{-1} - 7^{-1}) = 0$, while

$$\|c_j\|_2^2 = 2|\alpha_j|^2\varphi(q). \tag{10.16}$$

Consequently the normalized cross inner product is $1/2$, which proves (10.9).

On the other hand,

$$\sum_{u\in U_j} \zeta_q^u = 0, \tag{10.17}$$

so the orthogonal projection P_0 onto the constant vectors satisfies

$$P_0 c_j = \alpha_j \mathbf{1}, \qquad \|P_0 c_j\|_2^2 = \frac{1}{2}\|c_j\|_2^2. \tag{10.18}$$

Every multiplicative permutation fixes constant vectors. Hence (10.10) follows. Finally, (10.9) and the unchanged nonzero constant component show that $\mathcal{A}_j^* c_j$ is not a scalar multiple of c_j. There therefore exists a vector T orthogonal to c_j but not to $\mathcal{A}_j^* c_j$. For that T, the right side of (10.11) is zero and the left side is nonzero. Thus the displayed pointwise inequality cannot hold for arbitrary T. □

Remark 10.4 (The missing sesquilinear diagonal). For a finite prime set $\mathcal{P}$, put

$$z_a(T) = \langle \Pi_{a,j} T, c_j \rangle \qquad (a \in \mathcal{P}). \tag{10.19}$$

The square of the averaged coherent contribution is the sesquilinear expression

$$\left| \frac{1}{|\mathcal{P}|} \sum_{a \in \mathcal{P}} z_a(T) \right|^2 = \frac{1}{|\mathcal{P}|^2} \sum_{a,b \in \mathcal{P}} z_a(T)\overline{z_b(T)}. \tag{10.20}$$

Its loss relative to the diagonal branch energy is

$$\mathcal{V}_{\mathcal{P},j}(T) = \frac{1}{|\mathcal{P}|} \sum_{a \in \mathcal{P}} |z_a(T)|^2 - \left| \frac{1}{|\mathcal{P}|} \sum_{a \in \mathcal{P}} z_a(T) \right|^2 \tag{10.21}$$

$$= \frac{1}{2|\mathcal{P}|^2} \sum_{a,b \in \mathcal{P}} |z_a(T) - z_b(T)|^2. \tag{10.22}$$

A genuinely iterable arithmetic contraction would require a quantitative lower bound for (10.21) on the actual arithmetic channel vectors, together with a transfer identity showing that the corresponding diagonal branch energy reproduces the same family at the next step. Neither conclusion follows from (10.9). Thus the missing input is a closed sesquilinear diagonal estimate, not another explicit linear formula or another finite p-adic decomposition.

10.3. **Arithmetic iteration with fresh prime pairs.** We now pass from the finite permutation operator to an operator on arithmetic sequences. This logarithmic Liouville-correlation analysis is logically separate from the finite-height zero criteria proved above.

Let $\lambda(n) = (-1)^{\Omega(n)}$ be the Liouville function and remove its local factor at 3 by defining

$$f(n) = \lambda(n)(-1)^{v_3(n)} = \lambda\left(\frac{n}{3^{v_3(n)}}\right). \tag{10.23}$$

Then f is completely multiplicative, $f(3) = 1$, and

$$f(p) = -1 \qquad (p \neq 3 \text{ prime}). \tag{10.24}$$

Put

$$\eta_3(n) = (-1)^{v_3(n)+v_3(n+1)}, \tag{10.25}$$

$$b_3(n) = f(n)f(n+1). \tag{10.26}$$

Complete multiplicativity gives the exact factorization

$$\lambda(n)\lambda(n+1) = \eta_3(n) b_3(n). \tag{10.27}$$

On $\mathbb{Z}_3$ with normalized Haar measure, let $\mathcal{F}_k$ be the sigma-algebra of residue classes modulo 3^k, let

$$m_k = \mathbb{E}(\eta_3 \mid \mathcal{F}_k), \tag{10.28}$$

and define

$$\Delta_1 = m_1, \qquad \Delta_k = m_k - m_{k-1} \quad (k \geq 2). \tag{10.29}$$

The centered martingale decomposition and its energies are

$$\eta_3 = \sum_{k\geq 1} \Delta_k, \tag{10.30}$$

$$E_1 = \|\Delta_1\|^2_{L^2(\mathbb{Z}_3)} = \frac{1}{2}, \tag{10.31}$$

$$E_k = \|\Delta_k\|^2_{L^2(\mathbb{Z}_3)} = 3^{1-k} \qquad (k \geq 2). \tag{10.32}$$

For each layer define

$$F_k(n) = f(n+1)\Delta_k(n). \tag{10.33}$$

For complex sequences G_1, G_2, set

$$\mathcal{H}_X = \sum_{n\leq X} \frac{1}{n}, \tag{10.34}$$

$$\langle G_1, G_2\rangle_{\log,X} = \frac{1}{\mathcal{H}_X} \sum_{n\leq X} \frac{G_1(n)\overline{G_2(n)}}{n}, \tag{10.35}$$

$$\|G\|^2_{\log,X} = \langle G, G\rangle_{\log,X}. \tag{10.36}$$

For every positive integer a, let

$$(U_a G)(n) = G(an). \tag{10.37}$$

Since $|f| = 1$ and multiplication by an integer coprime to 3 permutes the triadic residue classes,

$$\|U_a F_k\|^2_{\log,X} \longrightarrow E_k \qquad (X \to \infty, 3 \nmid a) \tag{10.38}$$

for every fixed a and k.

Lemma 10.5 (Decorrelations of distinct arithmetic branches)**.** *For every fixed layer k and every pair of distinct positive integers a, b with $3 \nmid ab$,*

$$\langle U_a F_k, U_b F_k\rangle_{\log,X} \longrightarrow 0 \qquad (X \to \infty). \tag{10.39}$$

Proof. The product $\Delta_k(an)\Delta_k(bn)$ is periodic modulo 3^k. On a fixed residue class $n = 3^k t + s$, the two multiplicative factors are evaluated at

$$f(a3^k t + as + 1), \qquad f(b3^k t + bs + 1). \tag{10.40}$$

The determinant of these affine forms is

$$(a3^k)(bs+1) - (b3^k)(as+1) = 3^k(a-b) \neq 0. \tag{10.41}$$

The function f differs from λ at only the prime 3, so it satisfies the same nonpretentiousness condition. Tao's logarithmically averaged Elliott theorem, specifically Corollary 5 of [3], applies to the two fixed nonproportional affine forms. Replacing $(3^k t + s)^{-1}$ by $(3^k t)^{-1}$ creates an absolutely summable error. Summation over the finitely many residue classes proves (10.39). □

Choose pairs of primes

$$(p_1, q_1), (p_2, q_2), \ldots, \tag{10.42}$$

such that all the primes are mutually distinct and none equals 3. Define

$$B_s = \frac{1}{2}(U_{p_s} + U_{q_s}), \tag{10.43}$$

$$G_{r,k} = B_r \cdots B_1 F_k. \tag{10.44}$$

For $\varepsilon = (\varepsilon_1, \dots, \varepsilon_r) \in \{0,1\}^r$, put

$$m_\varepsilon = \prod_{s=1}^{r} \begin{cases} p_s, & \varepsilon_s = 0, \\ q_s, & \varepsilon_s = 1. \end{cases} \tag{10.45}$$

Then

$$G_{r,k} = 2^{-r} \sum_{\varepsilon \in \{0,1\}^r} U_{m_\varepsilon} F_k. \tag{10.46}$$

Unique factorization gives

$$m_\varepsilon = m_{\varepsilon'} \quad \Longleftrightarrow \quad \varepsilon = \varepsilon'. \tag{10.47}$$

Theorem 10.6 (Exact logarithmic contraction with fresh prime pairs)**.** *For every fixed $r, k \geq 1$,*

$$\lim_{X \to \infty} \|G_{r,k}\|_{\log,X}^2 = 2^{-r} E_k, \tag{10.48}$$

$$\lim_{X \to \infty} \|G_{r,k}\|_{\log,X} = 2^{-r/2} \sqrt{E_k}. \tag{10.49}$$

Thus the arithmetic contraction is iterable toward zero after the logarithmic limit is taken at each fixed depth.

Proof. Expanding (10.46) gives

$$\|G_{r,k}\|_{\log,X}^2 = 4^{-r} \sum_{\varepsilon, \varepsilon' \in \{0,1\}^r} \langle U_{m_\varepsilon} F_k, U_{m_{\varepsilon'}} F_k \rangle_{\log,X}. \tag{10.50}$$

The 2^r diagonal terms tend to E_k by (10.38). Every off-diagonal term tends to zero by Lemma 10.5 and (10.47). Hence the limit of (10.50) is $4^{-r} 2^r E_k = 2^{-r} E_k$, proving both assertions. □

Corollary 10.7 (Contraction in one active degree)**.** *To obtain a strict iterable contraction it is enough to fix one degree k with $E_k > 0$. For that single degree and every $r \geq 1$,*

$$\lim_{X \to \infty} \frac{\|G_{r,k}\|_{\log,X}^2}{\|F_k\|_{\log,X}^2} = 2^{-r}. \tag{10.51}$$

In particular, neither cancellation of the complete shell expansion nor the ordinary Cesaro form of Chowla is required for the decisive one-degree contraction lemma.

Proof. Divide (10.48) by $\lim_{X \to \infty} \|F_k\|_{\log,X}^2 = E_k > 0$. □

Proposition 10.8 (Noninvertibility boundary of the contraction)**.** *For every fixed active degree k, the decay of the transferred iterates does not, by itself, imply that the original layer vanishes. More precisely,*

$$\lim_{X \to \infty} \|F_k\|_{\log,X}^2 = E_k > 0, \tag{10.52}$$

$$\lim_{r \to \infty} \lim_{X \to \infty} \|G_{r,k}\|_{\log,X}^2 = \lim_{r \to \infty} 2^{-r} E_k = 0. \tag{10.53}$$

Consequently,

$$\boxed{\lim_{r \to \infty} \lim_{X \to \infty} \|G_{r,k}\|_{\log,X} = 0 \quad \not\Longrightarrow \quad F_k = 0.} \tag{10.54}$$

Thus the averaging operators B_s lose information, so decay alone cannot be read backward. Subsection 10.4 determines the exact scope of lower bounds involving the transferred energy.

Proof. The first identity is (10.38) with $a = 1$. The second is Theorem 10.6 followed by $r \to \infty$. Since $E_k > 0$, the same nonzero original layer coexists with arbitrarily small transferred iterates, proving (10.54). □

10.4. **Depth-uniform and finite-depth defect bounds.** The exact decay in Theorem 10.6 determines the effect of the depth quantifier: a fixed positive quantity cannot be bounded above by the same geometrically decaying channel energy at every depth.

Proposition 10.9 (Impossibility for a positive fixed atom [Unconditional])**.** *Fix a distinct positive ordinate τ, a degree $k \geq 1$, and constants $x > 0$, $x \neq 1$, and $\eta > 0$. If*

$$\eta D_x(\tau) \leq \liminf_{X\to\infty} \|G_{r,k}\|^2_{\log,X} \qquad (r \geq 0), \tag{10.55}$$

then $D_x(\tau) = 0$. Equivalently, if $D_x(\tau) > 0$, no fixed pair (k, η) can satisfy (10.55) *simultaneously for all transfer depths.*

Proof. Theorem 10.6 makes the right side of (10.55) equal to $2^{-r}E_k$. Therefore

$$0 \leq \eta D_x(\tau) \leq 2^{-r}E_k \qquad (r \geq 0). \tag{10.56}$$

Letting $r \to \infty$ gives $\eta D_x(\tau) = 0$, and $\eta > 0$ gives the conclusion. □

Corollary 10.10 (Impossibility of all-depth domination for a positive atom)**.** *If $D_x(\tau) > 0$, then*

$$\boxed{\forall k \geq 1\ \forall \eta > 0\ \exists r \geq 0 : \quad \eta D_x(\tau) > 2^{-r}E_k.} \tag{10.57}$$

Consequently, no fixed degree and no fixed positive coupling can dominate the same positive atom through all transfer depths.

Proof. For fixed k and η, choose any integer

$$r > \log_2\left(\frac{E_k}{\eta D_x(\tau)}\right). \tag{10.58}$$

Then $2^{-r}E_k < \eta D_x(\tau)$. The exact channel identity identifies $2^{-r}E_k$ with the limiting transferred energy. □

Proposition 10.11 (Finite-depth channel estimate [Unconditional reduction])**.** *Fix one depth $R \geq 0$, one degree $k \geq 1$, and one prescribed coupling $\eta > 0$. The finite-depth inequality*

$$\eta D_x(\tau) \leq \liminf_{X\to\infty} \|G_{R,k}\|^2_{\log,X} \tag{10.59}$$

is equivalent to the quantitative bound

$$D_x(\tau) \leq \frac{2^{-R}E_k}{\eta}. \tag{10.60}$$

At a single fixed depth, (10.59) *is compatible with $D_x(\tau) > 0$ and yields the explicit upper bound* (10.60)*.*

Proof. Apply Theorem 10.6 at the single fixed depth R and divide by $\eta > 0$. □

Remark 10.12 (Finite-depth scope)*.* Proposition 10.11 makes a finite quantitative assertion and does not assume persistence to infinite depth. A closed transfer identity showing that a prescribed coupling propagates from depth r to depth $r + 1$ would extend this estimate through the iterated channel. The contraction theorem alone gives no such propagation of a zero defect. Such an identity would be a sufficient mechanism, not a condition known to be necessary.

10.5. **Alternative exact routes and the channel–curvature boundary.** The second derivative is not the only exact finite-height route. Without introducing any further observable, the results already proved give two alternatives: equality $L_H = M_H$ on a set with an accumulation point, and equality $\nu_{H,x} = \mu_{H,x}$ at one fixed $x \neq 1$. Each is equivalent to RH(H). The curvature equality $F_H''(0) = 0$ is a third equivalent route.

Proposition 10.13 (What the present channels do not calculate [Unconditional audit])**.** *The identities proved in the arithmetic section do not give a formula for $F_H''(0)$ in terms of F_k, $G_{r,k}$, their shell energies, or their transferred resonances. In particular, the contraction*

$$\lim_{X\to\infty} \|G_{r,k}\|_{\log,X}^2 = 2^{-r}E_k \tag{10.61}$$

does not evaluate or force the vanishing of

$$F_H''(0) = \frac{2}{N(H)} \sum_{\substack{\rho\in\mathcal{Z}_H \\ \Re\rho>1/2}} \left(\Re\rho - \frac{1}{2}\right)^2. \tag{10.62}$$

Proof. Equation (10.62) is Theorem 8.2 and contains the horizontal zero coordinates. The definitions of F_k, $G_{r,k}$, E_k, and the transferred resonances contain the Liouville function, triadic shells, dilations, and logarithmic averages, but no height H, no horizontal zero coordinate, and no proved identity coupling these data to (10.62). Moreover, (10.52) and (10.53) show inside the arithmetic system itself that transferred decay coexists with positive source energy. Hence none of the displayed arithmetic identities yields the claimed curvature evaluation or vanishing. This proves the stated limitation of the present framework; it does not assert that no additional identity could ever be discovered. □

Remark 10.14 (Status of a possible future bridge). An independently proved identity or inequality that genuinely implies any one of the three equivalent finite-height equalities would be sufficient to prove RH(H). Such a particular arithmetic mechanism is not known to be necessary: the interval, atomic, and curvature criteria already provide different equivalent targets. Accordingly, no unproved channel-to-zero coupling is labelled "necessary" in this article.

Proposition 10.15 (Sesquilinear variance at every arithmetic level)**.** *Let*

$$G = G_{r-1,k}, \qquad V_p = U_{p_r}G, \qquad V_q = U_{q_r}G. \tag{10.63}$$

Define the two-branch variance

$$\mathcal{V}_r(X) = \frac{\|V_p\|_{\log,X}^2 + \|V_q\|_{\log,X}^2}{2} - \left\|\frac{V_p+V_q}{2}\right\|_{\log,X}^2. \tag{10.64}$$

Then

$$\mathcal{V}_r(X) = \frac{1}{4}\|V_p - V_q\|_{\log,X}^2, \tag{10.65}$$

$$\lim_{X\to\infty} \mathcal{V}_r(X) = \frac{1}{2}\lim_{X\to\infty} \frac{\|V_p\|_{\log,X}^2 + \|V_q\|_{\log,X}^2}{2}. \tag{10.66}$$

Thus the relative variance is $\delta = 1/2$ at every fixed level.

Proof. The first identity is the parallelogram law. The branch slopes in V_p have the form $p_r m_\varepsilon$, while those in V_q have the form $q_r m_{\varepsilon'}$. They are all distinct by unique factorization, so Lemma 10.5 gives

$$\langle V_p, V_q\rangle_{\log,X} \longrightarrow 0. \tag{10.67}$$

Theorem 10.6 gives

$$\|V_p\|_{\log,X}^2,\ \|V_q\|_{\log,X}^2 \longrightarrow 2^{-(r-1)}E_k. \tag{10.68}$$

Substitution in (10.64) proves (10.66). □

Theorem 10.16 (Closure and contraction of transferred triadic resonances)**.** *For all fixed $r, j, k \geq 1$,*

$$\limsup_{X\to\infty} \left| 2^{-r} \sum_{\varepsilon\in\{0,1\}^r} \langle b_3\Delta_j, U_{m_\varepsilon}(b_3\Delta_k)\rangle_{\log,X} \right| \leq 2^{-r/2}\sqrt{E_jE_k}. \tag{10.69}$$

If

$$\mathcal{R}_r(X) = 2^{-r} \sum_{\varepsilon\in\{0,1\}^r} \langle b_3\eta_3, U_{m_\varepsilon}(b_3\eta_3)\rangle_{\log,X}\,, \tag{10.70}$$

then

$$\limsup_{X\to\infty} |\mathcal{R}_r(X)| \leq 2^{-r/2} \left(\sum_{k\geq 1} \sqrt{E_k} \right)^2 \longrightarrow 0 \qquad (r\to\infty). \tag{10.71}$$

Proof. For $3 \nmid m$, complete multiplicativity and $f(n)^2 = 1$ give

$$b_3(n)b_3(mn) = f(m)f(n+1)f(mn+1). \tag{10.72}$$

Consequently,

$$b_3(n)b_3(mn)\Delta_j(n)\Delta_k(mn) = f(m)F_j(n)F_k(mn). \tag{10.73}$$

Every m_ε contains exactly r primes distinct from 3, and hence

$$f(m_\varepsilon) = (-1)^r. \tag{10.74}$$

It follows that the average inside (10.69) equals

$$(-1)^r \langle F_j, G_{r,k}\rangle_{\log,X}. \tag{10.75}$$

Cauchy–Schwarz, (10.38), and Theorem 10.6 prove (10.69).

Finally,

$$\sum_{k\geq 1} \sqrt{E_k} = \frac{1}{\sqrt{2}} + \sum_{k\geq 2} 3^{(1-k)/2} < \infty. \tag{10.76}$$

Expanding $\eta_3 = \sum_k \Delta_k$, applying (10.69), and summing the absolutely convergent majorant over j, k proves (10.71). □

10.6. **A single diagonal scale for all fixed layers.** The preceding fresh-prime theorem first fixes the depth and then lets X tend to infinity. A separate diagonalization produces one slowly growing prime scale that works for every fixed shell and for the complete summable shell expansion.

For $Q \geq 5$, define

$$w_\ell = \frac{\log \ell}{\ell}, \tag{10.77}$$

$$W(Q) = \sum_{\substack{\ell\leq Q \\ \ell \text{ prime},\ \ell\neq 3}} w_\ell, \tag{10.78}$$

$$P_Q = \frac{1}{W(Q)} \sum_{\substack{\ell\leq Q \\ \ell \text{ prime},\ \ell\neq 3}} w_\ell U_\ell, \tag{10.79}$$

$$D(Q) = \frac{1}{W(Q)^2} \sum_{\substack{\ell\leq Q \\ \ell \text{ prime},\ \ell\neq 3}} w_\ell^2. \tag{10.80}$$

The standard estimates $W(Q) \asymp \log Q$ and $\sum_\ell w_\ell^2 < \infty$ give

$$D(Q) \ll \frac{1}{(\log Q)^2}. \tag{10.81}$$

Theorem 10.17 (One slowly growing scale for all fixed layers)**.** *There exists a nondecreasing step function $Q(X) \to \infty$ satisfying*

$$\log Q(X) = o(\log X) \tag{10.82}$$

such that, for every fixed $k \geq 1$,

$$\|P_{Q(X)} F_k\|_{\log, X} \longrightarrow 0. \tag{10.83}$$

Proof. Choose a strictly increasing sequence $Q_r \to \infty$ such that

$$D(Q_r) < 2^{-4r}. \tag{10.84}$$

For fixed r, the family

$$\{\langle U_\ell F_k, U_q F_k\rangle_{\log, X} : k \leq r,\ \ell, q \leq Q_r,\ \ell \neq q\} \tag{10.85}$$

is finite. Lemma 10.5 therefore allows one to choose a strictly increasing sequence $X_r \to \infty$ so that, for every $X \geq X_r$ and every $k \leq r$,

$$\frac{1}{W(Q_r)^2} \sum_{\substack{\ell, q \leq Q_r \\ \ell, q \text{ prime},\ \ell, q \neq 3 \\ \ell \neq q}} w_\ell w_q \, |\langle U_\ell F_k, U_q F_k\rangle_{\log, X}| < 2^{-2r}. \tag{10.86}$$

By enlarging X_r if necessary, one may also require

$$\max_{\substack{k \leq r \\ \ell \leq Q_r,\ \ell \text{ prime},\ \ell \neq 3}} \left| \|U_\ell F_k\|_{\log, X}^2 - E_k \right| < 2^{-2r} \qquad (X \geq X_r), \tag{10.87}$$

$$\frac{\log Q_r}{\log X_r} < 2^{-r}. \tag{10.88}$$

Define

$$Q(X) = Q_r \qquad (X_r \leq X < X_{r+1}). \tag{10.89}$$

Then (10.88) implies (10.82). Expanding the squared norm of (10.79) gives, whenever $r \geq k$ and $X_r \leq X < X_{r+1}$,

$$\|P_{Q(X)} F_k\|_{\log, X}^2 \leq (E_k + 2^{-2r}) 2^{-4r} + 2^{-2r}. \tag{10.90}$$

The right side tends to zero, proving (10.83). □

Remark 10.18 (Endpoint correction). Uniformly for $\ell \leq Q(X)$,

$$\frac{1}{\mathcal{H}_X} \sum_{X/\ell < n \leq X} \frac{1}{n} \leq \frac{\log \ell + O(1)}{\log X} \tag{10.91}$$

$$\leq \frac{\log Q(X) + O(1)}{\log X} = o(1). \tag{10.92}$$

Thus the cutoff $n \leq X$ and the natural dilation cutoff $\ell n \leq X$ yield the same limit along this scale.

Theorem 10.19 (Single-scale contraction of the complete triadic expansion)**.** *Let*

$$F(n) = f(n+1)\eta_3(n) = \sum_{k \geq 1} F_k(n). \tag{10.93}$$

For the scale in Theorem 10.17,

$$\|P_{Q(X)} F\|_{\log, X} \longrightarrow 0. \tag{10.94}$$

Moreover,

$$\sum_{j,k\geq 1} \langle F_j, P_{Q(X)} F_k\rangle_{\log,X} \longrightarrow 0, \tag{10.95}$$

where the double series is interpreted through rectangular partial sums.

Proof. The shell norms are summable. For $J \geq 1$,

$$\sum_{k>J} \sqrt{E_k} = \sum_{k>J} 3^{(1-k)/2} = \frac{3^{-J/2}}{1-3^{-1/2}}. \tag{10.96}$$

Minkowski's inequality, permutation of triadic residue classes by $\ell \neq 3$, and (10.92) give

$$\limsup_{X\to\infty} \left\| P_{Q(X)} \sum_{k>J} F_k \right\|_{\log,X} \leq \sum_{k>J} \sqrt{E_k}. \tag{10.97}$$

For fixed J, Theorem 10.17 gives

$$\sum_{k\leq J} \|P_{Q(X)} F_k\|_{\log,X} \longrightarrow 0. \tag{10.98}$$

First let $X \to \infty$ and then $J \to \infty$ to obtain (10.94). Finally,

$$\left|\langle F, P_{Q(X)} F\rangle_{\log,X}\right| \leq \|F\|_{\log,X} \|P_{Q(X)} F\|_{\log,X} \longrightarrow 0. \tag{10.99}$$

The summable majorant in (10.96) permits expansion in rectangular partial sums and proves (10.95). □

Question 10.20 (Prescribed-scale bilinear target)**.** For a prescribed growing scale $Q = Q(X)$, can one prove, for each fixed k,

$$\left| \frac{1}{W(Q)^2} \sum_{\substack{\ell,q\leq Q \\ \ell,q \text{ prime},\ \ell,q\neq 3 \\ \ell\neq q}} w_\ell w_q \langle U_\ell F_k, U_q F_k\rangle_{\log,X} \right| = o(1)? \tag{10.100}$$

This is an averaged bilinear estimate over the two prime towers; it does not require a uniform estimate for every individual pair (ℓ, q). The polynomial scale $Q = X^\theta$ is a particularly strong instance not supplied by the diagonal argument.

Warning 10.21 (Polynomial scales and endpoints). For fixed $\theta > 0$, the choice $Q = X^\theta$ gives

$$\frac{\log Q}{\log X} = \theta, \tag{10.101}$$

not $o(1)$. Thus (10.100) would control the interior off-diagonal average, but it would not by itself identify $n \leq X$ with the natural cutoff $\ell n \leq X$. A polynomial-scale proof must retain the natural cutoff, control the endpoint separately, or subsequently send $\theta \to 0$.

Warning 10.22 (Exact scope of the contraction). The finite operator in Proposition 10.3 remains noniterable because of its invariant coefficient component. Theorem 10.6 and Corollary 10.7 overcome that particular obstruction after passing to the full arithmetic sequences and using a fresh prime pair at every level. They establish a stable sesquilinear contraction on any one fixed active degree. It is not necessary to prove the ordinary Cesaro form of Chowla or to control every shell simultaneously in order to prove that arithmetic contraction theorem.

Remark 10.23. No Landau formula or uniform variant is used in the finite-height zero criteria. Those proofs depend only on the finite zero symmetries, positivity of $\cosh u - 1$, the identity theorem, and a finite Vandermonde determinant. The separate arithmetic analysis in Subsection 10.3 additionally uses Tao's logarithmically averaged Elliott theorem.

11. Conclusion

For a fixed height H with $N(H) > 0$, the normalized complete-zero transform L_H and the ordinate-only transform M_H differ by an explicitly positive atomic defect localized at the distinct ordinates. The scalar difference is

$$L_H(x) - M_H(x) = \frac{2}{N(H)} \sum_{0<\tau\le H} D_x(\tau) e^{i\tau \log x}, \tag{11.1}$$

where each $D_x(\tau)$ is nonnegative and, for $x \neq 1$, vanishes precisely when the zeros at that ordinate lie on the critical line.

After the logarithmic substitution $x = e^y$, the defect is not used in the misleading form "a y-dependent coefficient times $e^{i\tau y}$." Expanding the hyperbolic cosine yields the genuine constant-coefficient exponential polynomial

$$F_H(y) = \frac{1}{N(H)} \sum_{\tau,\delta>0} m_H(\delta,\tau) \left[e^{(\delta+i\tau)y} + e^{(-\delta+i\tau)y} - 2e^{i\tau y} \right]. \tag{11.2}$$

Its uniqueness implies that equality of L_H and M_H on any set with an accumulation point forces every horizontal displacement δ to vanish.

The local expansion at $y = 0$ gives a still sharper characterization. The value and first derivative vanish automatically by symmetry, while

$$F_H''(0) = \frac{2}{N(H)} \sum_{\substack{\rho\in\mathcal{Z}_H \\ \Re\rho>1/2}} \left(\Re\rho - \frac{1}{2} \right)^2. \tag{11.3}$$

Thus finite-height critical-line containment is equivalent to a single positive curvature being zero. Exhausting the zeros through any unbounded sequence of heights converts each finite criterion into an equivalent formulation of the Riemann Hypothesis.

The triadic arithmetic analysis supplies a further ingredient. For any one fixed active shell k, a fresh pair of primes at each level produces the exact logarithmic contraction

$$\lim_{X\to\infty} \|G_{r,k}\|^2_{\log,X} = 2^{-r} E_k, \tag{11.4}$$

and the associated sesquilinear variance has the uniform relative value $1/2$. This is a genuine unconditional arithmetic theorem. It proves contraction of the transferred channel, not disappearance of its positive source energy and not a statement about the horizontal coordinates of zeta zeros. Ordinary Cesaro Chowla cancellation and simultaneous contraction of every degree are not prerequisites for this arithmetic lemma.

The exact status of the conditions in this article is consequently as follows.

(1) For fixed H, interval or accumulation-set equality of L_H and M_H, one-point equality of the atomic measures at any fixed $x \neq 1$, and the curvature equality $F_H''(0) = 0$ are each necessary and sufficient for $\mathrm{RH}(H)$. Along an unbounded sequence of heights, each is necessary and sufficient for the full Riemann Hypothesis.

(2) The arithmetic channel contraction (11.4) is unconditional, but it does not close the zero-side criteria by itself. The missing step is an explicit identity coupling a channel quantity to one of the three equivalent finite-height equalities.

(3) An identity connecting the existing branches or channels to any of the three exact zero-side equalities would provide a sufficient route. No particular identity of this kind is shown to be necessary.

In particular, the second derivative cannot presently be calculated from the displayed branch and channel identities. Its exact formula (11.3) uses the horizontal zero coordinates, whereas the arithmetic identities contain no proved coupling to those coordinates. The interval and atomic-measure equivalences are rigorous alternatives to the derivative criterion. Thus the arithmetic contraction is a complete result on the channel side, while the coupling identity required to close the zero-side criteria remains the precise outstanding relation.

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